Tichu

Statement

You have N cards in your hand. $N - K$ cards have integers on them. The remaining K cards are wild cards that can each represent a card of any integer you choose.

A **run** of length m is a sequence of m cards with the integers $(a, a+1, a+2, \ldots, a+m-1)$ for some integer a. For example, the sequences $(7,8,9)$, $(1,2,3,4,5)$, and (6) are runs, but $(2,3,5,6)$, $(1,6)$, and $(4,3,2,1)$ are not runs.

What is the longest **run** you can make with these cards? You may rearrange the cards in your hand however you like to form this run.

Input Format

- The first line consists of an integer N $(1 \le N \le 10^5)$, followed by another integer K $(0 \le K \le N)$.
- The second line contains $N - K$ integers, representing the cards in your hand that aren't wild cards: $C_1, C_2, \ldots, C_{N-K}$ $(1 \le C_i \le 10^9)$.

Output Format

Output a single integer, the length of the longest run you can form from the cards you have.

Sample Input

```
11 2
2 5 4 3 3 8 7 11 15
```

Sample Output

```
8
```

Explanation

If you make the wildcards a 6 and a 9, you can make the run $(2,3,4,5,[6],7,8,[9])$, which has length 8. You cannot make a longer run.